Intention To Treat (ITT)

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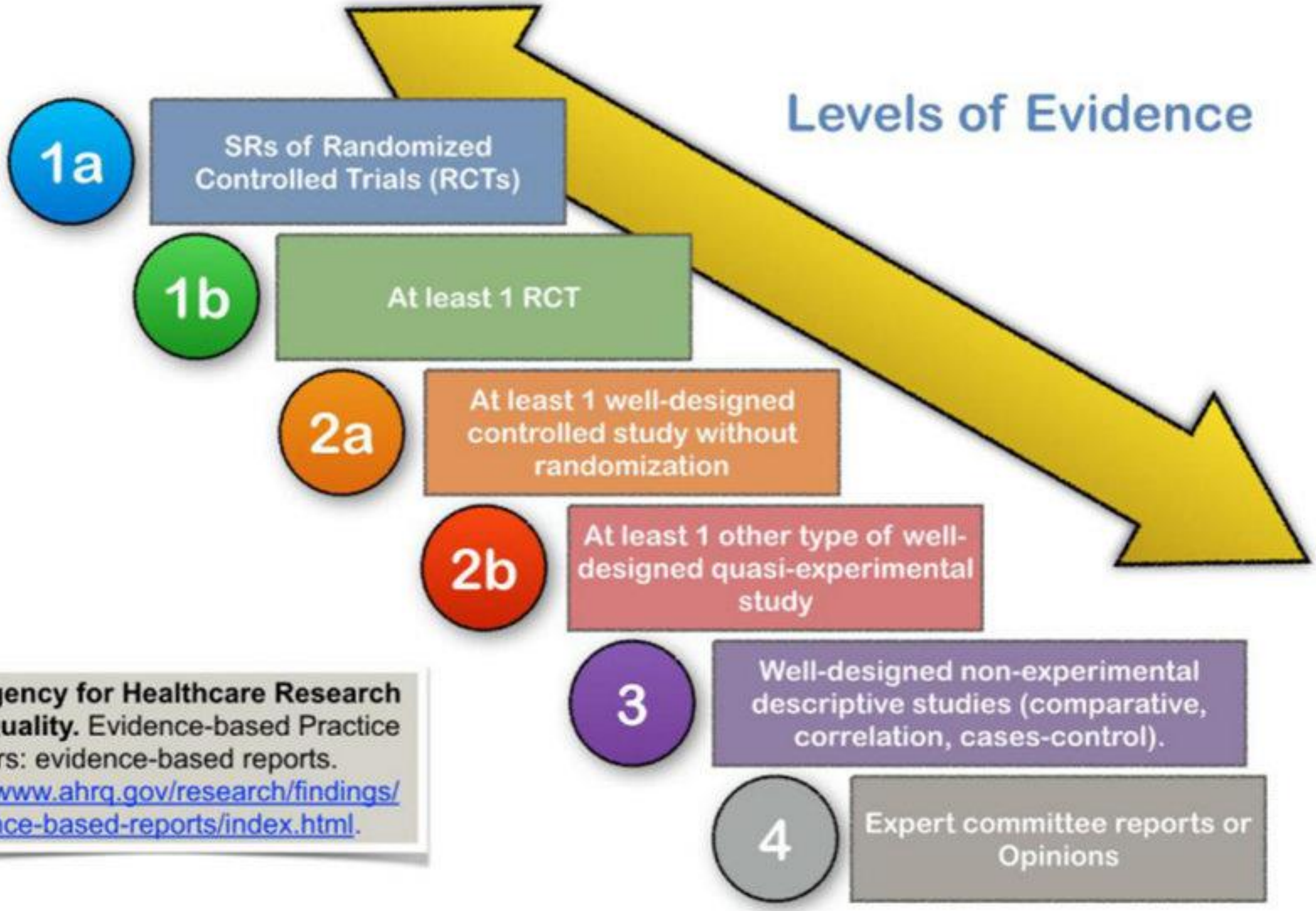
Disclosures

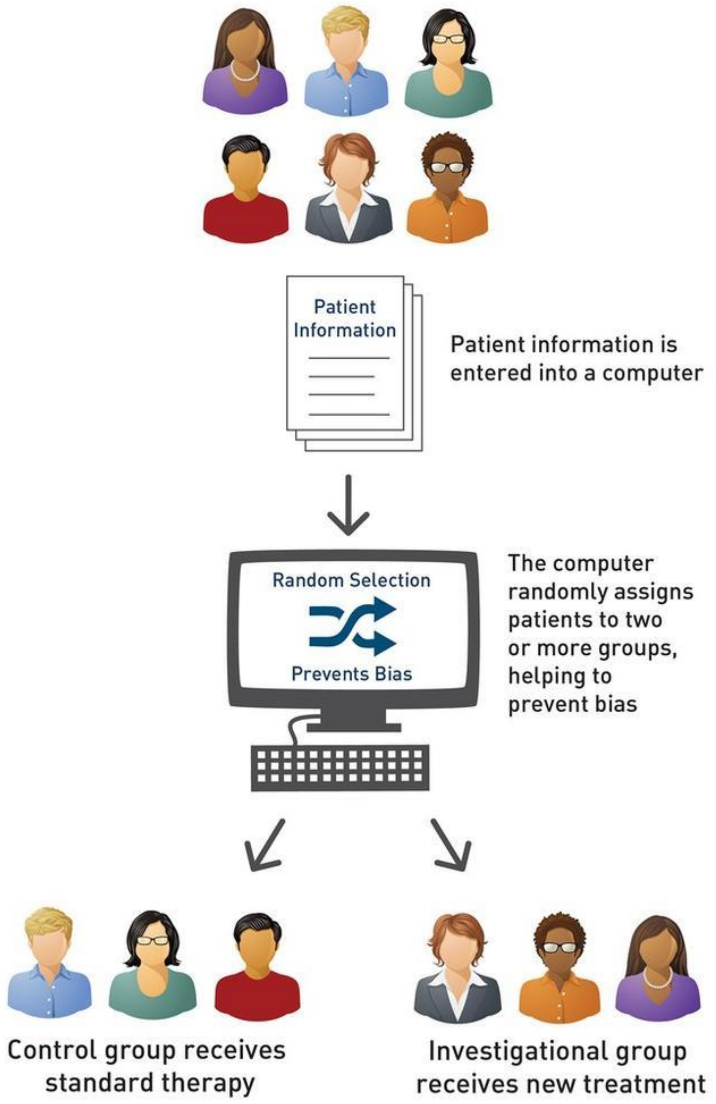
- NIH Grants
  - NS065257
  - GM113852
- Industry:
  - Eli Lilly (statistical/research consultant)
- Statistical Editor
  - *Anesthesiology*
  - *Headache*

Levels of Evidence



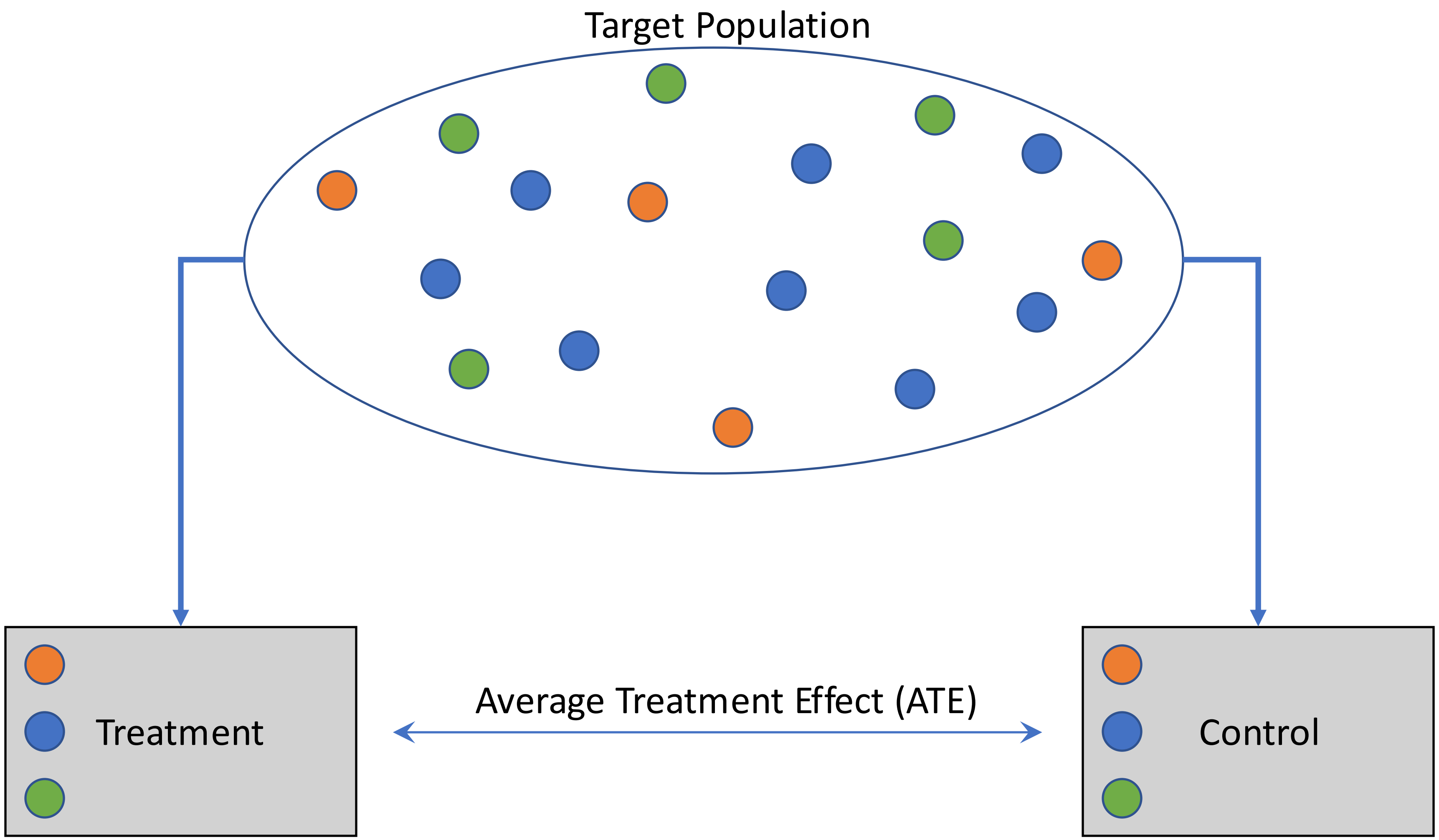
US Agency for Healthcare Research and Quality. Evidence-based Practice Centers: evidence-based reports. http://www.ahrq.gov/research/findings/evidence-based-reports/index.html.

CLINICAL TRIALS RANDOMIZATION



Randomized Controlled Trial

$E(X_{tx}) = E(X_{con})$



$E(Y_{TX}) – E(Y_{CON})$ = Treatment Effect

The Problem

- In idealized RCTs, everyone is ‘fully observed’
  - Baseline characteristics are measured
  - Treatment provided as intended
  - Participant does not change their mind about participation
  - Outcome measures successfully collected

- In reality, there are MANY reasons why a participant my not be ‘fully observed’
  - This challenges the core design of an RCT

Advice: The next time you read a RCT, attend to these reasons

Intercurrent Events

- Post-randomization event (or experience) that impacts the measurement of treatment effect
- Causes each individual to experience their own ‘path’ through the trial
  - Creates difficulty in interpretation

- Examples:
  - Finishes study exactly as protocolized
  - Starts alternative treatment (or palliative treatment)
  - Develops a side-effect that requires monitoring or intervention
  - Discontinues treatment
  - Death (for studies where death is not an outcome)
  - Rescue therapy used
  - Undergoes additional surgery/intubation

The Problem: Bias

- Bias occurs when an estimated treatment effect differs from its ‘true’ value
  - Example: the treatment does not work as well as expected when given to the target population

- Several types of bias in RCTs
  - Selection bias
  - Performance bias
  - Detection bias
  - Attrition bias

https://catalogofbias.org/

Intention to Treat (ITT)

- A strategy for reducing the risk of bias (e.g., attrition/selection) in the estimation of a treatment effect

- Simple in principle: Analyze the the treatment arms as they were randomized. Regardless of:
  - Compliance/adherence
  - Cross-over
  - Attrition

Analyze as Randomized!

Benefits of Intention to Treat (ITT)

- Helps preserve prognostic balance
- Reduces chances of artificial selection of inferences by subgroups (i.e., completers)
- Emphasizes accountability by study team
  - Everyone randomized will be included in the analysis!
- Enhances generalizability (external validity) of the findings

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1124168/

Criticism of Intention to Treat (ITT)

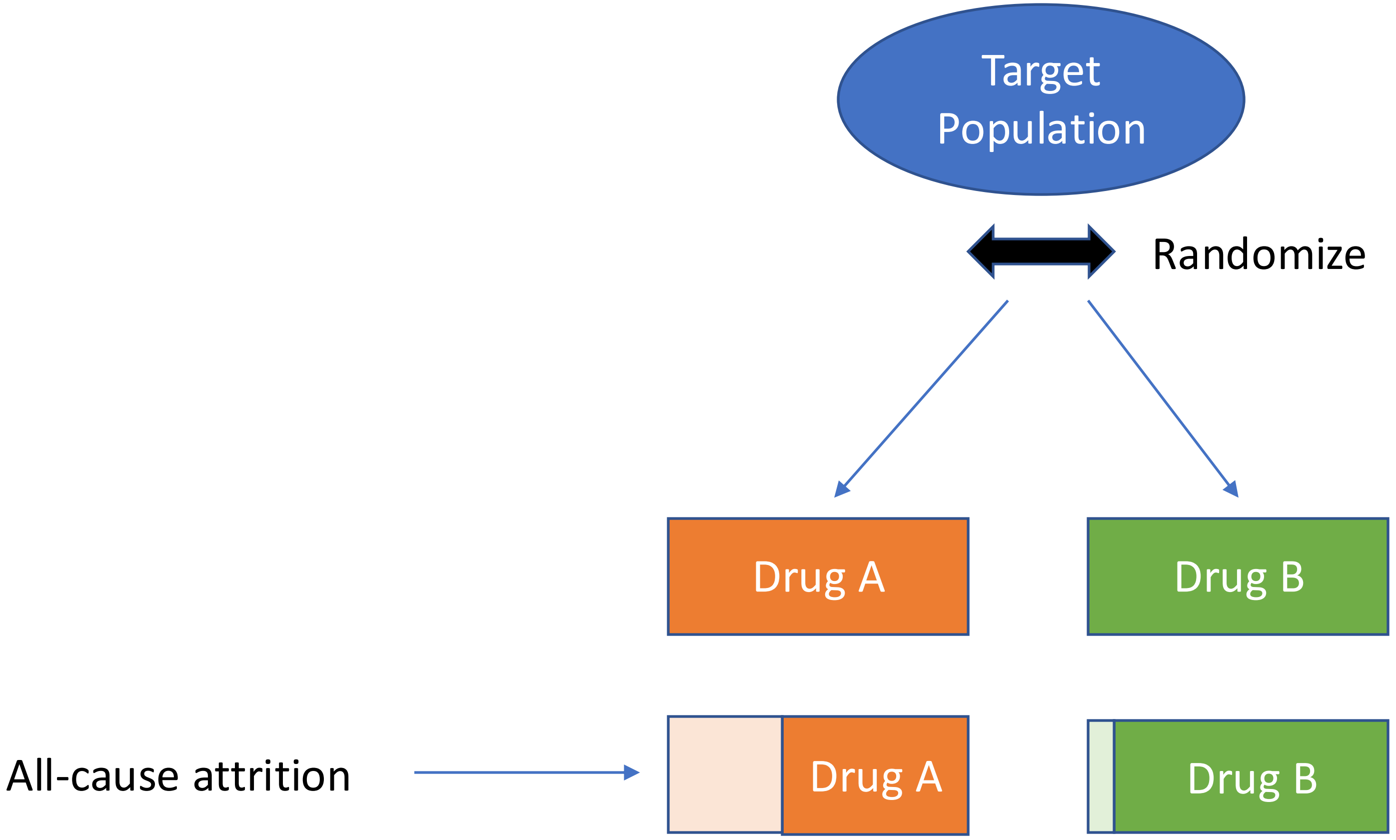
- Too cautious and results in:
  - Lower statistical power (ie., individuals are included in the analysis who were not actually treated)
  - Less likely to show a positive treatment effect

- Most RCTs intend to demonstrate efficacy NOT effectiveness
  - This should involve examining the actual treatment effect when given under intended conditions
  - Per protocol (PP) is closer to the idealized objectives of a clinical trial

Modified ITT

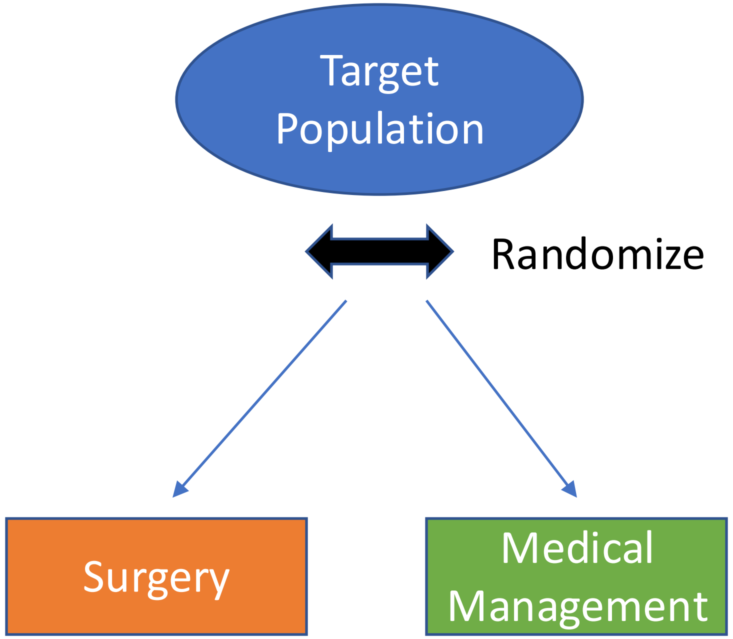
Famous Examples of Intention to Treat: 1

- An RCT is examining the benefit of two drugs on a nasty outcome.



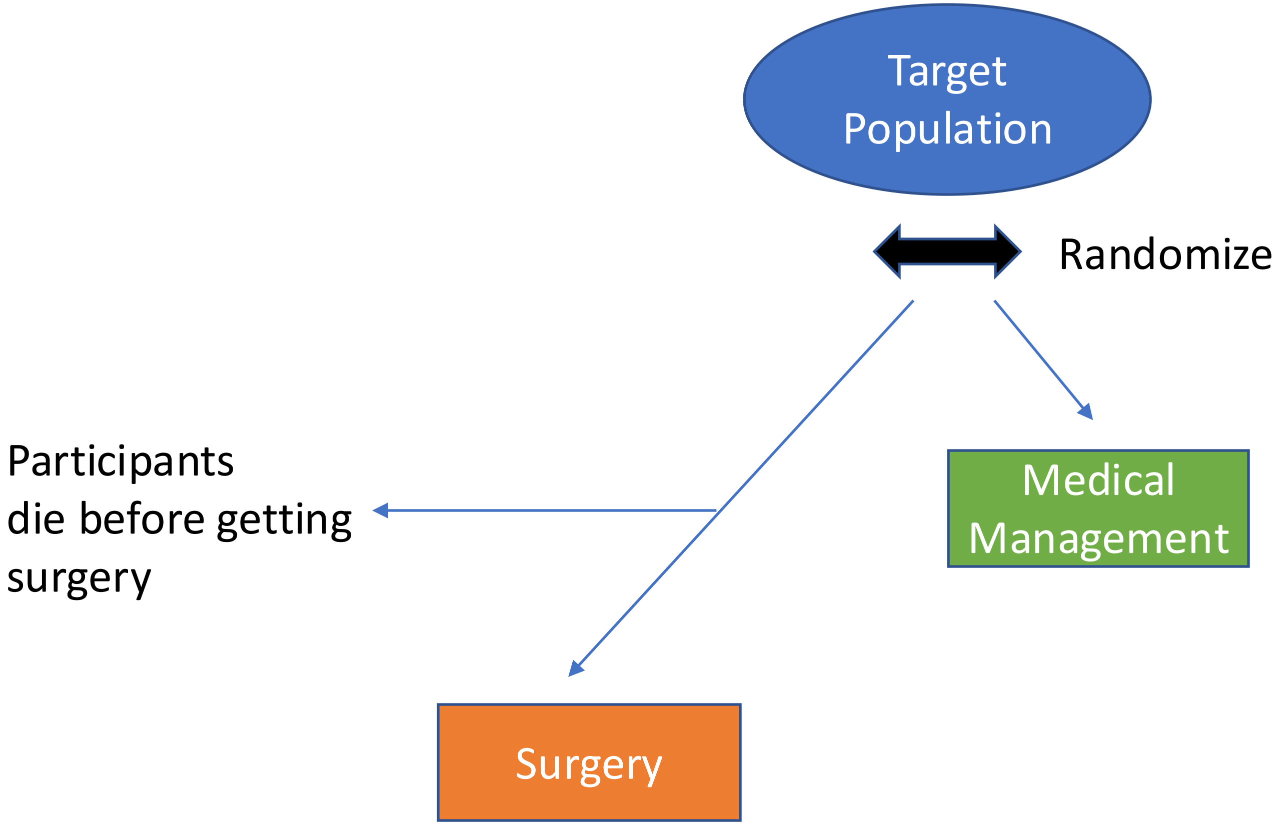
Famous Examples of Intention to Treat: 2

- An RCT is examining the benefit of surgery on a nasty outcome.



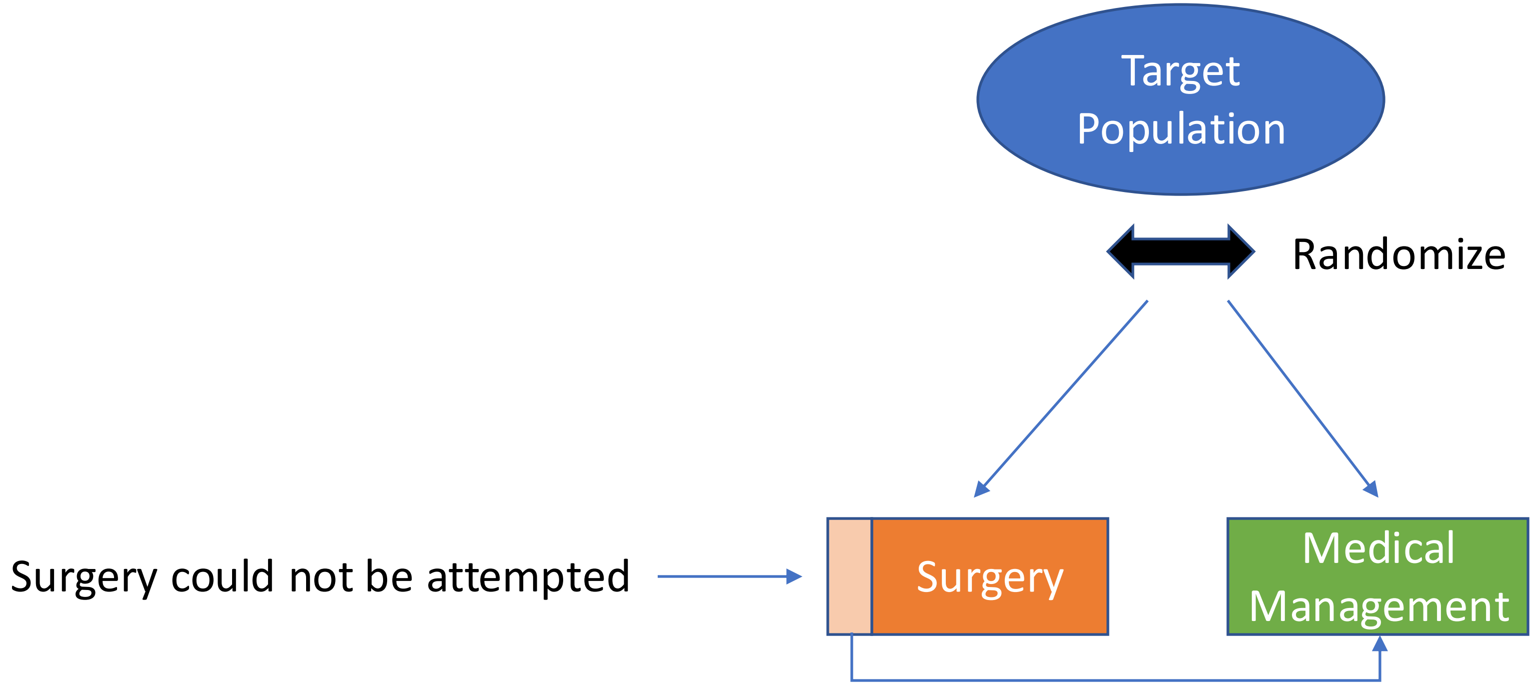
Famous Examples of Intention to Treat: 2

- An RCT is examining the benefit of surgery on a nasty outcome.



Famous Examples of Intention to Treat: 3

- An RCT is examining the benefit of surgery on a nasty outcome.



Honoring Intention to Treat (ITT) Philosophy

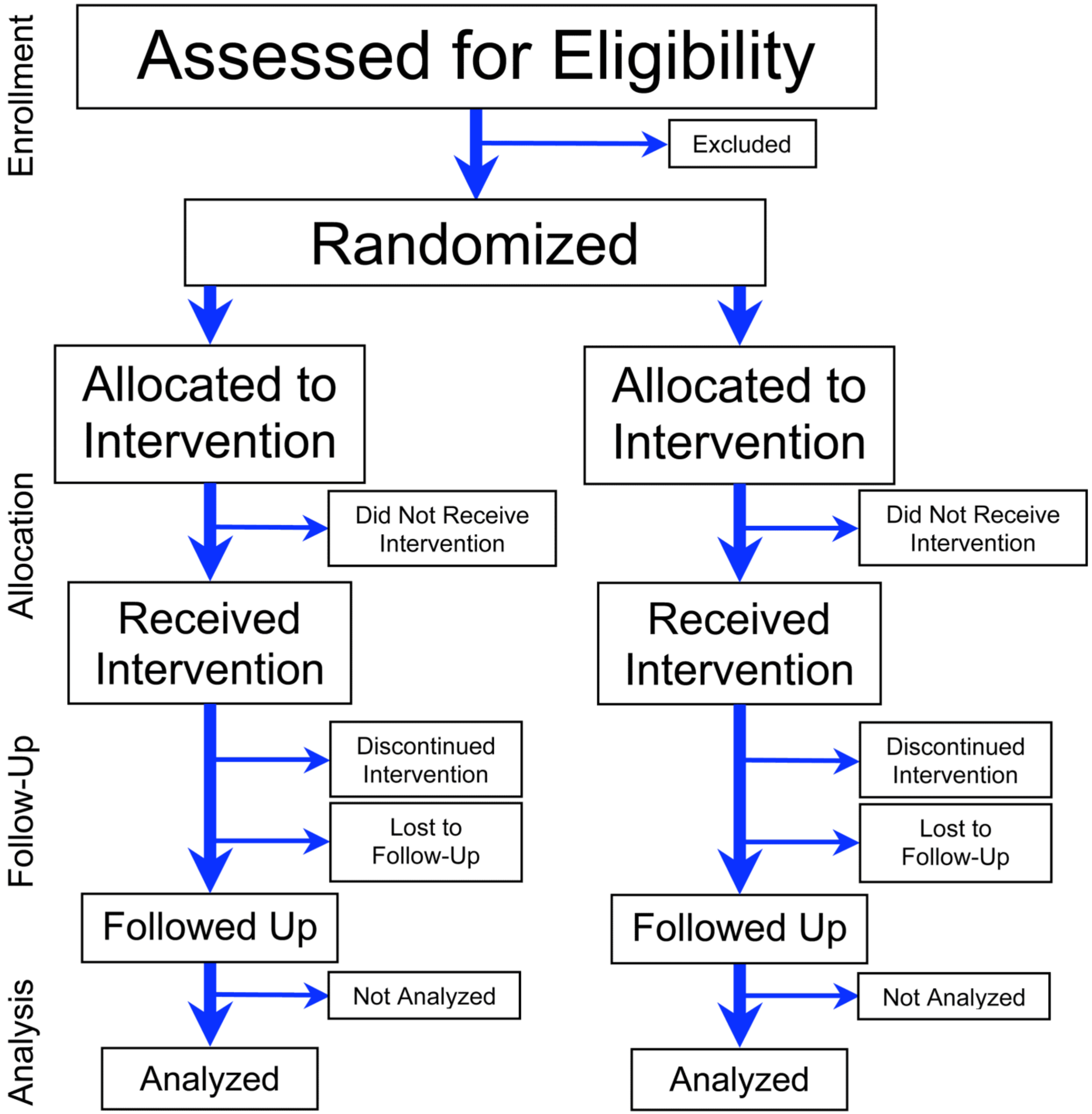
- It is <u>NOT</u> sufficient to merely state that ITT was used
  - This is unfortunately common
  - "We enrolled 45 individuals to get 40 valid patients…"
- Must have a strategy to enact ITT methods
  - Find some way to include individuals in the analysis according to their intended treatment arm
- Intention to Treat is usually an adventure in missing data

Intention to Treat (ITT) strategies

- For ‘fully observed’ participants
  - Code them in the treatment arm they were randomized to
    - Example: A participant is coded as received Drug A even if they wrongly received Drug B

- Missing data requires imputation
  - Worst case scenario
    - Example: missing participants assumed to have experienced the outcome
  - Last observation carried forward (LOCF)
  - Single imputation
  - Multiple imputation
  - Maximum Likelihood estimation

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3651592/



Enrollment
Assessed for Eligibility
Excluded
Randomized
Allocation
Allocated to Intervention
Did Not Receive Intervention
Received Intervention
Allocated to Intervention
Did Not Receive Intervention
Received Intervention
Follow-Up
Discontinued Intervention
Lost to Follow-Up
Followed Up
Discontinued Intervention
Lost to Follow-Up
Followed Up
Analysis
Not Analyzed
Analyzed
Not Analyzed
Analyzed

Estimands: A Brief Note

- Addendum to the ICH E9 in 2017
  - Reflects the current thinking of the FDA

- Estimand
  - A precise definition of the treatment effect being estimated

- Estimator
  - How the treatment effect will be calculated (estimated)

- Estimate
  - The resulting value used to interpret the treatment effect

Four Elements of the Estimand

- The population
  - Precise definition of eligibility including dynamic changes to eligibility
- The variable(s)
  - Precise definition of outcome and timing of measurement(s)
- Intercurrent events
  - Events (experiences) that occur after intervention (randomization?)
- Population-level summary
  - Precise definition of the effect estimate, its form, and method

Strategies for Intercurrent Events

- Treatment Policing Strategy
  - Value of the outcome regardless of intercurrent events
  - ITT
- Composite Strategy
  - Intercurrent event woven into the outcome
  - “Proportion with delirium according to … and not reintubated or experiencing additional surgery…”
- Hypothetical Strategy
  - Assumes that the intercurrent event did not happen (needs a statistical model)
  - “Proportion who experienced or would have experienced delirium had they been able to test…”
- Principal Stratum Strategy
  - Considers measurements in subgroup(s) of the target population (by design)
  - Dynamic randomization (by former intercurrent event)
- While on Treatment Strategy
  - Evaluate treatment effect prior to the occurrence of the intercurrent event
  - Measurements stop after intercurrent event

Thanks!

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